

AquaNet: Gravito-Quantum Network

A Symbiotic Framework for Entangled Phase-Lock Communication

Overview

AquaNet is a next-generation decentralized network protocol built on the principles of **Gravito-Quantum Mechanics**. Unlike classical networks that rely on bit-stream transmission over physical infrastructure, AquaNet utilizes **Entangled Phase-Locking** between localized quantum clusters.

By leveraging the **SPHY Formalism**, AquaNet treats data as a phase-state within a gravitational field, enabling near-instantaneous synchronization (E-Ping) that is immune to classical interception or decryption.

Key Components

- **Fermion Identifiers (UID)**: Structural rigid points of the network that provide deterministic identity.
- **Boson Force-Fields**: Information carriers that fluctuate according to the $B(t)$ phase-field.
- **Anyon Braiding**: The topological link that creates the entanglement "bridge" between two bodies.

System Requirements

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### Hardware & Environment
- OS: Pop!_OS / Ubuntu / FreeBSD (Optimized for Unix-like kernels)
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- Language: Rust (Edition 2021) & Python 3.10+
- Libraries (Rust): num-complex, sha2, serde_json, cargo
- Libraries (Python): py5 (Processing for Python), pandas, pyarrow (Parquet support), numpy
- Computing: Multi-core CPU for quantum state simulation; GPU support recommended for 3D visualization.

What You See in the Emulator

The **AquaNet Visualizer** provides a real-time tactical view of the quantum state between two clusters (AQUA-A and AQUA-B):

- **The Two Bodies:** You are observing two distinct particle clouds. These are the localized nodes of the network.
- **Dynamic Fermions (Gold):** The vibrating core points represent the rigid identity (UID) of the clusters. Their "jitter" simulates Pauli exclusion and thermal stability.
- **Pulsating Bosons (Cyan):** These points represent the signal field. They expand and contract following the *B(t) Magnitude*. When the field is cyan/bright, the system is reaching coherence.
- **The E-Ping Bridge (Magenta):** When the *RESONANCE* exceeds the critical threshold (4.7+), a magenta braiding appears between the clusters. This is the **Topological Bridge**. It proves the bodies are entangled, allowing secure packet delivery.
- **HUD Metrics:** Large indicators track the *B(t) Magnitude* and *Coherence State*. If the state drops to 'DECOHERENCE', the bridge vanishes, illustrating how the network naturally "evaporates" packets to prevent eavesdropping.

Usage

1. Run the Generator: `python sphy_aquanet_dual_body_gen.py`
2. Launch the Emulator: `python sphy_dual_body_animated_v2.py`